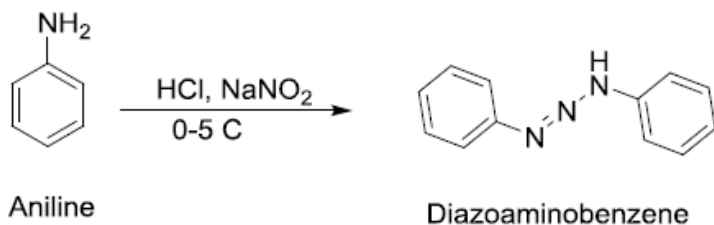


Experiment-3

Aim: Preparation of Diazoamino Benzene and report the melting point and yield of the product.

Reaction:



1. Chemical/Materials:

Aniline 9 mL
Conc. HCl: 14 mL
Sodium Nitrite 3.5 gm
Sodium Acetate: 14 gm
Distilled Water: 250 mL

2. Apparatus Required:

1. Beaker 250 mL (No: 2).
2. Conical Flask 100 mL & 250 mL
3. Small Glass Stirring Rod: (No: 1)
4. Spatula
5. Magnetic Stirrer with stir bar
6. Thermometer
7. Measuring Cylinder: 100 mL and 10 mL
8. 10 mL Glass Pipette with Suction Bulb
9. Buchner Funnel/ or Glass Stemless Funnel
10. Filter Paper

Procedure: Take 50 mL distilled water in a 250 mL Beaker. Stir (use stir bar and magnetic stirrer) it on water-ice bath and add 16 mL of Conc. HCl (0.2 moles) portion wise (2 – 3 mL at a time). Allow the solution to cool to 0-5 °C and then add 9.3 mL (0.1 moles) aniline using glass pipette & suction bulb. Add approx. 50g crushed ice into the reaction mixture with stirring. Add 0.05 moles of NaNO₂ solution (3.5 gm / 15 mL H₂O) dropwise into the reaction mixture with continuous stirring. Allow the reaction mixture to stir for another 15 minutes at 0-5 °C. Add 0.25 moles of sodium acetate solution (20 gm / 50 mL H₂O) slowly with stirring. Yellow precipitate will form, continue stirring for another 15 minutes. Then filter the yellow solid using buckner funnel with filter paper and wash the solid with distilled water. Dry the solid on whatmann filter paper in a petri dish in a Hot-Air Oven at 45-50 °C. Report the yield and melting point of dried solid of diazoamino benzene.

**Result:**

Yield

Melting Point [96-98 °C]

Precautions:

Apron, Safety Goggles and Gloves must wear during Experiment. Manual Stirring can be done with Glass Rod in place of magnetic stirrer. Recommended to conduct this experiment in chemical fume hood.
